

Asal Mehradfar

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EDUCATION

University of Southern California Ph.D., Electrical Engineering (Advisor: Prof. Salman Avestimehr)	Los Angeles, CA 2023 – Present
University of Southern California M.Sc., Electrical Engineering	Los Angeles, CA 2023 – 2025
Sharif University of Technology B.Sc., Electrical Engineering B.Sc., Computer Science	Tehran, Iran 2017 – 2022 2019 – 2022

RESEARCH EXPERIENCE

AI-Driven Analog and RF Circuit Design (Fall 2023 – Present)

- Developed **AICircuit**, a multi-topology benchmark for AI-driven analog and RF circuit design. [3], [7]
- Introduced **FALCON**, a layout-aware ML framework for automated analog circuit inverse design. [2]
- Designed graph-based circuit representations enabling GNN generalization across unseen topologies. [2]
- Developed EM-aware physical synthesis and inductor modeling techniques for RF circuits. [1]
- (Ongoing) Advancing toward frequency-aware optimization, automated layout generation, and LLM-based expert design systems.

AI-Driven Lipid Nanoparticle Design for Drug Delivery (Fall 2023 – Present)

- Developed **LANTERN**, an ML framework for predicting LNP transfection efficiency. [4]
- (Ongoing) Developing generative models for lipid design toward improved functional and physicochemical properties.
- (Ongoing) Advancing ML models for in-vivo biodistribution prediction of liver vs. non-liver targeting.

AI for Financial Systems (Fall 2024 – Present)

- Developed **CryptoMamba**, an SSM-based framework for cryptocurrency time-series forecasting. [5]
- (Ongoing) Authoring a survey on AI-driven cryptocurrency trading and financial modeling.

Additional Research Experience

- Developed **ATHENA**, a test-time steering framework for improving count fidelity in text-to-image diffusion models.
- Developed uncertainty-aware routing methods for LLMs to optimize cost–accuracy trade-offs. [6]
- Designed an action recognition and multi-camera tracking system for resource-constrained edge devices.

HONORS AND AWARDS

Qualcomm Innovation Fellowship Finalist	2025
Reproducibility Award at Machine Learning and the Physical Sciences Workshop @ NeurIPS	2024
Annenberg Graduate Fellowship @ USC	2023

PUBLICATIONS

- [1] Y. Huang, **A. Mehradfar**, S. Avestimehr, and H. Aghasi, "EM-Aware Physical Synthesis: Neural Inductor Modeling and Intelligent Placement & Routing for RF Circuits", *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2026.
- [2] **A. Mehradfar**, X. Zhao, Y. Huang, E. Ceyani, Y. Yang, S. Han, H. Aghasi, and S. Avestimehr, "FALCON: An ML Framework for Fully Automated Layout-Constrained Analog Circuit Design", *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- [3] **A. Mehradfar**, X. Zhao, Y. Niu, S. Babakniya, M. Alesheikh, H. Aghasi, and S. Avestimehr, "Supervised Learning for Analog and RF Circuit Design: Benchmarks and Comparative Insights", *arXiv preprint arXiv:2501.11839*, 2025.
- [4] **A. Mehradfar**^{*}, M. S. Sepehri^{*}, J. M. Hernandez-Lobato, G. S. Kwon, M. Soltanolkotabi, S. Avestimehr, and M. Rasoulianboroujeni, "LANTERN: A Machine Learning Framework for Lipid Nanoparticle Transfection Efficiency Prediction", *arXiv preprint arXiv:2507.03209*, 2025.
- [5] M. S. Sepehri^{*}, **A. Mehradfar**^{*}, M. Soltanolkotabi, and S. Avestimehr, "CryptoMamba: Leveraging State Space Models for Accurate Bitcoin Price Prediction", *IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 2025.
- [6] T. Zhang^{*}, **A. Mehradfar**^{*}, D. Dimitriadis, and S. Avestimehr, "Leveraging Uncertainty Estimation for Efficient LLM Routing", *Collaborative and Federated Agentic Workflows Workshop @ ICML*, 2025.
- [7] **A. Mehradfar**, X. Zhao, Y. Niu, S. Babakniya, M. Alesheikh, H. Aghasi, and S. Avestimehr, "AICircuit: A Multi-Level Dataset and Benchmark for AI-Driven Analog Integrated Circuit Design", *Machine Learning and the Physical Sciences Workshop @ NeurIPS*, 2024.

ACADEMIC SERVICE

- Reviewer, IEEE Transactions on VLSI Systems 2026
- Reviewer, AI for Science Workshop @ NeurIPS 2025
- Reviewer, ML4PS Workshop @ NeurIPS 2025, 2024

SKILLS

- **Programming:** Python, MATLAB, C/C++, Java
- **Machine Learning:** PyTorch, PyTorch Geometric, TensorFlow, scikit-learn
- **Tools:** $\LaTeX$, Git